

Lesson Plan

Course Code: CSE 3105

Course Title: Microprocessors and Embedded System

Timeline	Topics /contents	Learning Outcomes
Week 1 ,2	Introduction and overview of microprocessors	• Overview of microprocessors, number system • Introduction to assembly language, brief concept of compiler, assembler, linker and debugger • Evolution of computers • ROM and RAM families • Bus structure
Week 3, 4	Microcomputer Structure	• CPU modules and architecture, Bus configuration, CPU registers. • Interrupt and Interrupt Handling. • Read, Write operation, DMA
Week 5, 6	8085 and 8086 microprocessor	• Introduction to 8085 Architecture, Timing Diagram, Instruction Formats. • Introduction to 8086 Architecture, registers, pin diagram, physical address. • Instruction Formats, Instruction Set, Assembler Directives, Interrupts of 8086. • Features, Signals, I/O & Memory Interfacing
Week 7	Addressing modes, Coprocessors	• Memory addressing technique of 8086 • learn the examples of addressing modes • 8087 coprocessor structure
CT - 1		
Week 8, 9	Introduction to assembly language and machine language programming	• Basics of the assembly language program. • Branching structure. • Learn about jump, CMP, if-then. • Allocate assignment for the course
Week 10	Assembly language, Stacks, Procedures.	• Understand how looping structures work. • Analyze SHIFT and ROTATE instructions. • Understand the process of shifting and rotating.
CT - 2		
Week 11, 12	8051 microcontroller architecture	• Hardware features, Architecture, Internal RAM structure. • Special Function Registers, Memory Organization, I/O Ports and Circuits. • Timers, Interrupts, Serial Communication, Interfacing of External Memory. • interfacing LCD & Keyboard, Real Time Clock.
Week 13, 14	8051 programming, Peripheral devices.	• Understand Addressing Modes, Instruction Set of 8051, Assembly Language Programming and C Programming, -A/D & D/A Interface; • USART (8251), Interrupt Controller (8259) Keyboard and Display Controller.
CT - 3		